Algorithms and Programming II Lab

DEVELOPING A JAVA/PROCESSING VIDEOGAME | SECOND SEMESTER-2026

Objective

Develop a fully animated 2D video game in Processing/Java focusing on the healthy use of social networks.

Story

The video game must revolve around the healthy use of social networks, addressing aspects related to digital well-being, digital citizenship, time management, online safety, critical thinking, and the balance between digital life and real life. The proposal aims to develop educational video games that allow players to reflect on these topics through interactive experiences and decision-making.

Possible development lines include:

- Social network simulators
- Time management games
- Interactive stories
- Life simulators
- Cooperative games
- Adventure or platform games with mechanics related to digital well-being
- Adventure or platform games with mechanics related to the responsible use of technology

Game Mechanics

On-Screen Elements

- Avatar
- Moving floating messages
- Score indicator

Suggested Controls

- Keyboard or mouse movement
- **B Key:** Block incoming messages (if the game focuses on incoming and outgoing social media messages)
- **R Key:** Report negative message (if the game focuses on incoming social media messages)

- **C Key:** Capture positive messages (if the game focuses on incoming and outgoing social media messages)
- **P Key:** Pause

Game Rules

Each message that can be displayed must manage the following message types:

- **Negative messages (red):** Deduct well-being if they reach the avatar.
- **Positive messages (green):** Increase points and well-being.
- **Neutral messages (gray):** Do not affect stats, but cause distraction.

The player must:

- Intercept messages before they reach the avatar.
- Avoid blocking positive messages (incurs a penalty).

> **Healthy Usage Pattern System**
>
> If 3 negative messages appear in a row, the Report option is activated.
>
> Using the R key removes all negative messages on screen and awards bonus points.

Technical Requirements

Object-Oriented Programming (OOP)

- **Avatar Class:** Position, well-being, draw method.
- **Message Class:** Type, text, position, speed, move method, draw method, collision detection.
- **Game or System Class:** Game state, score management, message generation.

Use of Arrays

- One-dimensional array to store active messages.
- Optional: Two-dimensional array to represent a chat wall.

Minimum Conditions

- Start screen with instructions.
- Gameplay screen with moving messages.
- Score and well-being system.
- At least two types of messages.
- Game Over screen and restart option.

Final Educational Message

Upon completing the game, a reflective message regarding the proper use of social networks must be displayed.

Notes

- The setting or landscape, including all objects and the avatar, can be freely chosen.
- Speed control is autonomous; that is, each team designs their acceleration strategy for the messages.
- Teams may decide on the final scenario.
- It is essential to keep in mind that adding extra activities not explicitly included in the base requirements will aid in the process of selecting the three best games.
- The lab must apply the concepts of arrays and procedures.
- Development teams must consist of three students.

Submission Guidelines

FIRST DELIVERY (DUE TWO WEEKS AFTER FIRST EXAM)

Must include a report containing:

- Introduction (explaining the laboratory topic).
- Definition of three roles and who worked on each role for the first delivery (Back-end, Front-end, and Integrator).
- A video where each member explains how they performed their respective role (maximum 2 minutes in length).
- Initial designs or sketches of the video game.
- A sketch displaying the video game entrance/start screen.

Packaging: Both the report and video must be submitted in a single .zip file named team_#, where # represents the team number (e.g., team_13).

SECOND DELIVERY (FINAL DELIVERY)

Must include a report containing:

- Team presentation sheet.
- Introduction (explaining the laboratory topic).
- Minimum requirements to install and execute the video game.
- A section explaining how to install the video game.
- Main algorithms developed for the video game (not the raw programs).
- Conclusions (explaining your perception of the lab learning experience).
- A .zip file of the project developed in Java.

Packaging: Both the report and the project .zip must be delivered in a single master .zip file named team_#, where # is the team number (e.g., team_13).

- The delivery of this second lab will take place on the date stipulated in the syllabus (parceling) for the second lab delivery.
- The second report should build upon and expand the report submitted in the first delivery.